FEChAR Tutorial

The heating-ramp problem

Folder `Tutorials/ConvectiveRamp_Bprime` | Earth, Mars, Venus, Titan

What this tutorial does

FEChAR simulates how a heat-shield sample (here TACOT, a carbon-phenolic material) heats up and burns away. In the **ramp problem** the front face is heated by a short pulse: the heating rises to a plateau, stays there, then drops, and the sample cools.

You can run the same problem in four atmospheres, **Earth, Mars, Venus or Titan**, by changing one word.

The tutorial has three parts:

1. **Run** the simulation (Section).
2. **Post-process** the results with script, (Section):
3. **Change** the ramp or the heating level (Section).

The heating pulse

The heating at the edge of the boundary layer is given by two numbers: the mass-transfer coefficient $\rho_e u_e C_H$ [kg/m^2/s] and the recovery enthalpy H_r [J/kg]. Both follow the same pulse in time (Figure 1). The code uses the surface-chemistry tables (B' tables) of the chosen atmosphere to compute how fast the char burns away. The ramping problem is detailed in the Benchmark problem [2] presented in the Ablation workshop.

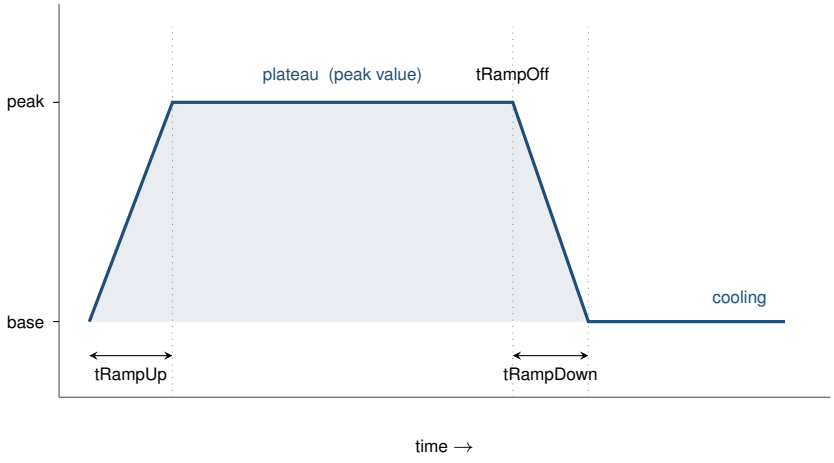


Figure 1. The heating pulse. Every heating quantity goes from its base value to its peak value, stays at the peak, then returns to the base value.

Table 1. Pulse settings (in `params.idp`).

Setting	Flag	Default	Meaning
`tRampUp`	`-tup`	0.1 s	time to rise to the plateau
`tRampOff`	`-toff`	60 s	time when the ramp-down starts (end of plateau)
`tRampDown`	`-tdown`	0.1 s	duration of the ramp-down
`reuechPeak`	–	0.3 kg/m^2/s	plateau of $\rho_e u_e C_H$
`heatbc`	`-heat`	high	plateau of H_r: low 1.5, intermediate 13.3, high 25 MJ/kg

The equations are in `docs/THEORY.md`; you do not need them to follow this tutorial.

Before you start

Software. FreeFEM++[1] (included in the official installers from freefem.org) on Linux, macOS or WSL. For post-processing: Python 3 with NumPy and Matplotlib.

Where to work. Always start from inside the tutorial folder. All paths are relative.

```
FEChAR/
  Datafolder/                  material model, B' tables, heating macros (shared)
  scripts/                     post-processing scripts (Python)
  Tutorials/ConvectiveRamp_Bprime/
    params.idp                 all settings: edit this file (or use command-line flags)
    main.edp                   the solver; normally not edited
    run_all_environments.sh    runs the four environments one after the other
```

Part 1: Run the simulation

Step 1. Open the folder.

```
cd FEChAR/Tutorials/ConvectiveRamp_Bprime
```

Step 2. Choose the environment. Set `Environment` in `params.idp`, or give `-env` on the command line. It selects the surface-chemistry table and the wall pressure.

Table 2. The wall pressure used for the four environments.

`-env`	Wall pressure
Earth	101 325 Pa
Mars	610 Pa
Venus	9.2×10^6 Pa
Titan	146 900 Pa

Step 3. Choose the heating level. `heatbc` (flag `-heat`) is `low` (1.5 MJ/kg), `intermediate` (13.3 MJ/kg) or `high` (25 MJ/kg).

Step 4. Run. Use the quick test first. The flag `-nw` switches off the graphics window. The current time is printed while the run advances.

```
# quick test: Mars, high heating
FreeFem++ -nw main.edp -env Mars -heat high -tend 1 -toff 4 -ele 250 -dt 0.005

# full run: uses the settings in params.idp
FreeFem++ -nw main.edp
```

Table 3. Quick test and full run.

Setting	Flag	Quick test	Full run (default)
end time	`-tend`	8 s	120 s
ramp-down starts	`-toff`	4 s	60 s
elements	`-ele`	100	500
time step	`-dt`	0.005 s	0.005 s

> **Keep dt $\leq$ 0.005 s.** A larger time step (0.01 s) made the run diverge in tests. For a faster test, reduce the end time and the number of elements, not the time step.

The results are written to a folder named `case_<Environment>_Bprime_<heat>`, for example `case_Mars_Bprime_high`. **Every run deletes and recreates this folder**, so copy results elsewhere if you want to keep them.

Part 2: Post-process the results

Once the simulation has finished, post-process the results by running `postprocess.py` from the tutorial folder. This single script automatically extracts temperature, pressure and density histories at fixed depths, determines the positions of the virgin and char interfaces, and generates the required plots.

Post-processing: extract data and generate plots

For a single case, run:

```
python ../../scripts/postprocess.py case_Earth_Bprime_high \
```

The script automatically performs the following operations:

1. Extracts the temperature, pressure and density histories at the specified depths and saves them as `probes\_temperature.da` `probes\_pressure.dat` and `probes\_density.dat` in each case folder.
2. Determines the positions of the virgin and char interfaces as functions of time and saves the results in `interfaceloc.dat` in each case folder.
3. Generates the wall-temperature, surface-recession, blowing, in-depth temperature-profile, interface-position and probe-history plots. The figures are saved in the specified output directory.

When multiple case folders are supplied, the relevant curves are overlaid in the same figures to facilitate comparison between environments. The extracted data files are saved in their respective case folders, while the figures are saved in the `figures` directory.

Thus, after each simulation has finished, a single command is sufficient to post-process the results and generate the required data files and plots.

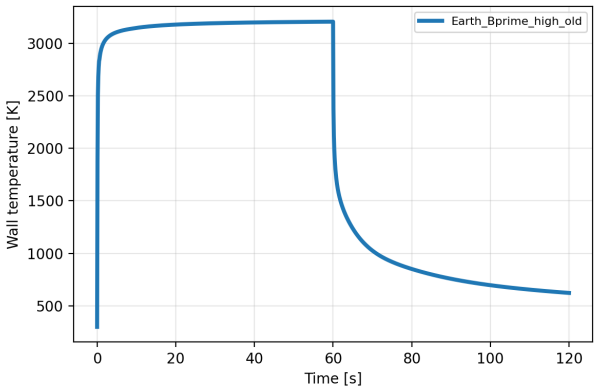


Figure 2. Wall temperature.

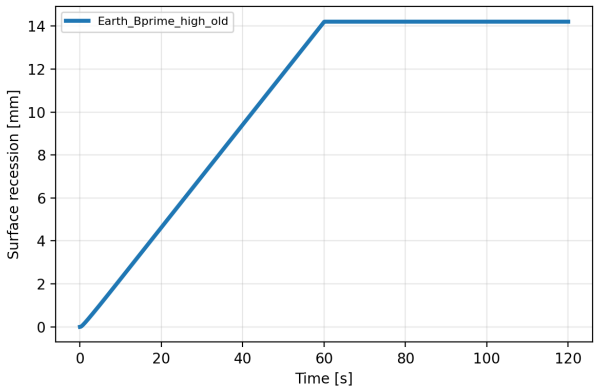


Figure 3. Surface recession.

What the simulation writes

The simulation output files are plain text, with columns containing the computed physical quantities. The post-processing script reads these files and generates the additional data and figures automatically. The main simulation output files are listed in Table 4.

Part 3: Change the ramp

Ramp times

Use the flags `-tup`, `-toff`, `-tdown`, or edit `tRampUp`, `tRampOff`, `tRampDown` in `params.idp`. Set `-tend` larger than `-toff` + `-tdown` if you want the cooling phase.

```
# slower ramp: 1 s ramp-up, plateau until 4 s, 1 s ramp-down, cooling until 8 s
FreeFem++ -nw main.edp -env Earth -tup 1 -toff 4 -tdown 1 -tend 8 -ele 100
```

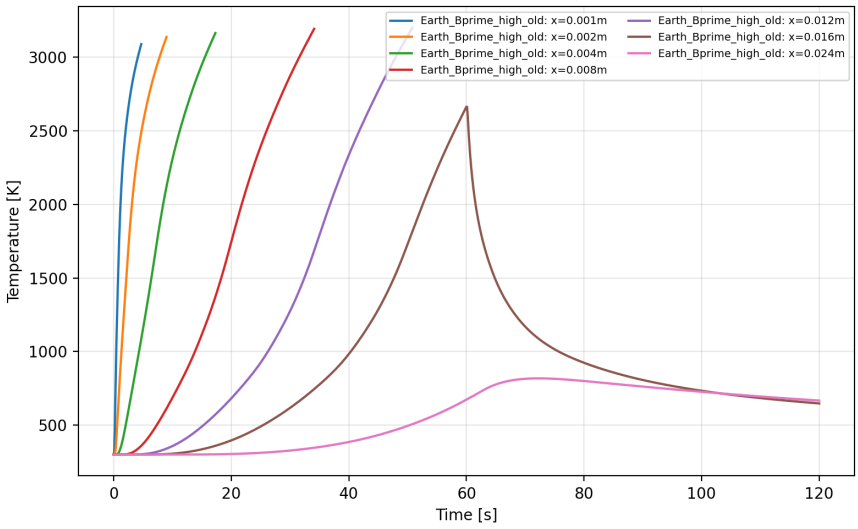


Figure 4. Temperature histories at fixed depths inside the material.

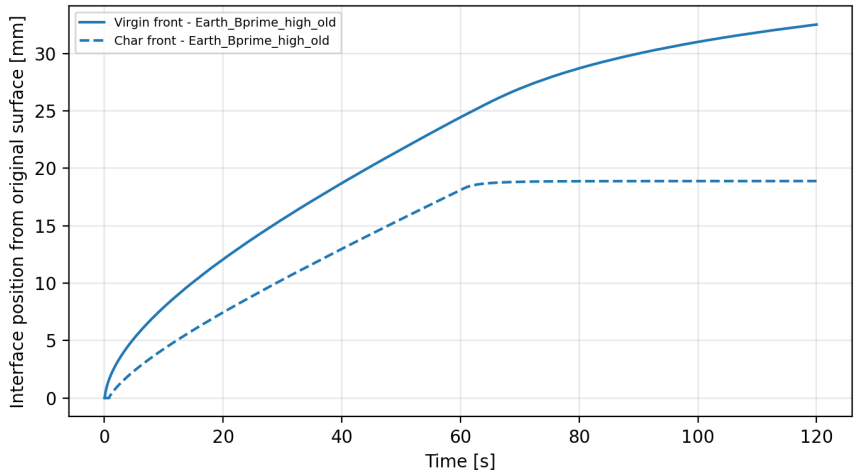


Figure 5. Positions of the virgin and char interfaces.

Ramp times of 0.1 s and longer were tested. For a faster ramp, reduce `-dt` and check that the run stays stable.

A new heating level

Add one line to the list in `params.idp` (section 2), then run with `-heat medium`:

```
if (heatbc == "low") HrPeak = 1.5e6;
else if (heatbc == "intermediate") HrPeak = 1.33e7;
else if (heatbc == "high") HrPeak = 2.5e7;
else if (heatbc == "medium") HrPeak = 8.0e6;   // new level [J/kg]
```

Going further

- **Different pulse shape:** replace the macro `pulse(t)` in `Datafolder/bc/Boundary1.idp`. It must return a number between 0 and 1 for every time (the file is shared by all three tutorials).
- **New environment:** add a table named `Bprime_table_modified_<Name>.dat` to `Datafolder/tacotdata/Bprimetables/`, add its wall pressure to `params.idp` (for example `else if (Environment == "MyPlanet") prwall = 5.0e4;`) and run with `-env MyPlanet`.
- **Tables off:** with `useAblation = false` the tables are not used and the sample does not recede. All environments then give almost the same result, so do not use this setting to compare environments.

Table 4. Files in the case folder.

File	Columns
`out_<t>.dat`	x [m] (from the original surface), T [K], pressure [Pa], density [kg/m^3]
`boundarydat.dat`	t, pressure at the front node, wall temperature T_w [K], B'_g, B'_c, h_w [J/kg]
`BC.dat`	t, recession rate [m/s], front-face position [m]
`mgeq.dat`	t, pyrolysis-gas mass flux at the wall, char mass flux at the wall [kg/m^2/s]
`WEnergyBalance.dat`	t, then in W/m^2: All the terms in the surface energy balace are compared
`inflocv.dat`, `inflocc.dat`	t, density threshold, position of the virgin front (98 % virgin) and of the char front (2 % char remaining)
`info.dat`	run header (environment, heating level, ramp times, settings) and final thickness

If something goes wrong

Message or symptom	What to do
`Load error : msh3`	FreeFEM has no `msh3` plugin. See “Ubuntu note” in Section 3.
`Error opening file ../../Datafolder/...`	You started FreeFEM outside the tutorial folder. `cd` into `Tutorials/ConvectiveRamp_Bprime` first.
`Sorry error in Optimization ...` or `nan`	The run diverged. Use `-dt 0.005` or smaller.
`ERROR: unknown Environment ...`	Misspelled name (capital letters matter), or it is not defined in `params.idp`.
A post-processing script cannot find the files	Check the case-folder name and that the run has finished. Run the scripts from the tutorial folder.
The run is very slow	Full runs take hours. Use the quick test first and keep `ploton = false`.

Quick reference

Table 5. Command-line flags. Each one overrides the matching line of `params.idp`.

Flag	Overrides	Default	Values
`-env`	`Environment`	Earth	Earth, Mars, Venus, Titan
`-heat`	`heatbc`	high	low, intermediate, high
`-tend`	`tend`	120	end time [s]
`-ele`	`ele`	500	number of elements
`-dt`	`dt`	0.005	time step [s]; keep ≤ 0.005
`-tup`	`tRampUp`	0.1	ramp-up time [s]
`-toff`	`tRampOff`	60	time when the ramp-down starts [s]
`-tdown`	`tRampDown`	0.1	ramp-down time [s]

References

[1] Frédéric Hecht. New development in freefem++. *Journal of Numerical Mathematics*, 20(3–4):1–4, 2012. https://doi.org/10.1515/jnum-2012-0013.

[2] Jean R Lachaud, Alexandre Martin, Tom van Eekelen, and Ioana Cozmuta. Ablation test-case series# 2. numerical simulation of ablative-material response: Code and model comparisons. 2012. https://jeanlachaud.com/?page_id=132.